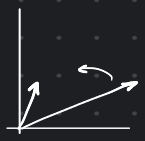
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Eigen-stuff



when we multiply a vector to a matrix, it gets rotated and/or scaled.

Eigen vector:
- Non-zero vector corresp. to a matrix s.t. the matrix only scales it without changing the direction.
  $Mv = \lambda v$ (λ : scalar)
  ↳ eigen value : scalar that tells the amount of scaling that happens to the corresp. eigenvector.
- Exist only for square matrices.

$Av = \lambda v$ (eigenvector, eigenvalue)

e.g. $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix} = 5\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

↳ eigenvector ↳ eigenvalue corresp. to $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

Significance/applications

- In case of differential calculus where matrices are involved.
- Google PageRank
- Weather prediction

Computation

$$Av = \lambda v$$

$$Av - \lambda v = 0$$

$$(A - \lambda I)v = 0$$

$$A = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$$

step 1: compute $A - \lambda I$.

$$A - \lambda I = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} = \begin{bmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{bmatrix}$$

step 2: solve $\det(A - \lambda I) = 0$ ← characteristic eqn

$$\begin{vmatrix} 4-\lambda & 1 \\ 2 & 3-\lambda \end{vmatrix} = 0$$

$$(\lambda - 4)(\lambda - 3) - 2 = 0$$

$$\lambda^2 - 7\lambda + 10 = 0$$

$$(\lambda - 2)(\lambda - 5) = 0$$

$$\lambda = 5, 2$$

Step: solve $(A - \lambda I)v = 0$

$\lambda_1 = 5$:

$$\begin{pmatrix} 4-5 & 1 \\ 2 & 3-5 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = 0$$

$$\begin{pmatrix} -1 & 1 \\ 2 & -2 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$\lambda_2 = 2$:

$$\begin{pmatrix} 4-2 & 1 \\ 2 & 3-2 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 2 & 1 \\ 2 & 1 \end{pmatrix}\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{pmatrix} 1 \\ -2 \end{pmatrix}$$

eigenvalues : 5, 2

eigenvectors : $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\begin{bmatrix} 1 \\ -2 \end{bmatrix}$

example:

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$|A - \lambda I| = 0$$
$$\Rightarrow (1-\lambda)(4-\lambda) - 4 = 0$$
$$\Rightarrow \lambda^2 - 5\lambda = 0$$
$$\Rightarrow \lambda = 0, 5$$

$\lambda = 0$:

$$\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
$$\left[\begin{array}{cc|c} 1 & 2 & 0 \\ 2 & 4 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 1 & 2 & 0 \\ 0 & 0 & 0 \end{array}\right]$$
$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = t \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

$\lambda = 5$:

$$\begin{bmatrix} -4 & 2 \\ 2 & -1 \end{bmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$
$$\left[\begin{array}{cc|c} -4 & 2 & 0 \\ 2 & -1 & 0 \end{array}\right] \rightarrow \left[\begin{array}{cc|c} 2 & -1 & 0 \\ 0 & 0 & 0 \end{array}\right]$$
$$\begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = t \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

If a matrix has zero eigenvalues:
- matrix is singular
- cols are LD, NOT full rank matrix.

NOTE: For an $n \times n$ matrix, we can have n diff. eigen values.

If a matrix has multiple zero eigen values (k), the matrix has k LD cols in A.

Properties

- Eigen values of diagonal matrix are the diagonal entries themselves.

$$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \rightarrow \begin{bmatrix} a-\lambda & 0 \\ 0 & b-\lambda \end{bmatrix} \qquad (a-\lambda)(b-\lambda) = 0$$
$$\lambda = a, b$$

- Sum of eigen values = trace(A) = sum of diagonal elements.

$$\sum_{i=1}^{n} \lambda_i = tr(A)$$

$\lambda_1, \lambda_2, \ldots, \lambda_n$ soln to $\rightarrow |A - \lambda I| = 0$

polynomial will be like $(-1)^n \left[\lambda^n + c_1 \lambda^{n-1} + \cdots + c_{n-1}\lambda + c_n\right] = 0$

- Product of eigenvalues = $\det(A)$
- Eigen values of symmetric matrices are real.
- Eigen values of skew symmetric matrices are imaginary.
- Eigen values of orthogonal matrices have absolute value 1.

Proof:

$$Av = \lambda v$$
$$\|Av\| = \|\lambda v\| = |\lambda| \, \|v\|$$

If A is orthogonal $\rightarrow \|Av\| = \|v\|$

$$\Rightarrow |\lambda| = 1$$

- Eigen values of Δas matrices are the diagonal entries.
- Scaling of eigen vectors.

An eigen vector scaled by a non-zero value still remains an eigenvector for the corresponding eigenvalue.

$$Av = \lambda v$$
$$\Rightarrow A(cv) = \lambda(cv)$$

- $A_{n \times n}$ has n diff eigenvalues. $\Rightarrow$ corresp. eigen vectors will act as the basis for $\mathbb{R}^n$.
- If λ is the eigenvalue for A, then λ^k is the eigenvalue for A^k for the same eigenvector v.

$$Av = \lambda v$$
$$A^2 v = \lambda A v$$
$$= \lambda \lambda v$$
$$A^2 v = \lambda^2 v$$
$$\vdots$$
$$A^k v = \lambda^k v$$

- Performing row operations might change λ.

Samved Bangale

- An $n \times n$ matrix can have max. n eigen values.
- corresp. to 1 λ, we can have multiple eig. vecs.
- If A has n diff. eig. values then eigen vectors form basis for $\mathbb{R}^n$.

e.g. $\begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \rightarrow \lambda = 2, 2$ (Dar)

$(A - \lambda I)v = 0$

$\left[\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right]$ $v_1 = s$, $v_2 = 0$ $\quad s\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

- corresp. to 1 eigen value, we can have atmost k LI eigen vectors if λ is repeated k times.

Arithmetic multiplicity (λ) = how many times λ is repeated.

Geometric multiplicity (λ) = # of LI eigen vectors corresp. to λ.

$GM(\lambda) \leq AM(\lambda)$

- The eigenvectors corresponding to different eigenvalues will be LI.

$\begin{bmatrix} 2 & 1 \\ 0 & 3 \end{bmatrix} \rightarrow \lambda = 2, 3$

$\lambda = 2$: $\left[\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 1 & 0 \end{array}\right] \rightarrow s\begin{bmatrix} 1 \\ 0 \end{bmatrix}$

$\lambda = 3$: $\left[\begin{array}{cc|c} -1 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right] \rightarrow s\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ } LI

$\begin{bmatrix} 0 & 2 & 3 \\ 0 & 2 & 4 \\ 0 & 0 & 3 \end{bmatrix}$ $\lambda = 0, 2, 3$

$\lambda = 0$: $\left[\begin{array}{ccc|c} 0 & 2 & 3 & 0 \\ 0 & 2 & 4 & 0 \\ 0 & 0 & 3 & 0 \end{array}\right] \rightarrow \left[\begin{array}{ccc|c} 0 & 2 & 3 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \rightarrow s\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$

- If $\lambda = 0$ and multiplicity = K:
  - → k LD col^ms in A.
  - → Dim. N(A) = k
  - → rank = (n-k)

Q. $\lambda(\lambda-1)(\lambda-2)(\lambda-4)^3$

→ # of eigen vectors = 6 → $A_{6\times 6}$

4 diff. eigen values (0, 1, 2, 3)

$\lambda = 4$ has multiplicity 3 → max. 3 LI eig. vectors

$\lambda = 0$ has multiplicity 1 → 1 LD col^m. nullity = 1, rank = 5

atleast 4 LI eig. vectors

atmost 6 LI eig. vectors

Q. $\lambda^6 (\lambda-3)^4$

→ $n = 10$ → $A_{10\times 10}$

$\lambda = 0$ has multiplicity 6 → max 6 LI eig. vecs.

$\lambda = 3$ multiplicity 4 → max 4 LI eig. vecs.

dim of null space

atleast 2 LI eig. vectors.

Q. $(\lambda-1)(\lambda-2)(\lambda-3)$

→ $n = 3$ $A_{3\times 3}$

$\lambda = 1, 2, 3$

Q. $\lambda^2 (\lambda-3)^2 = 0$

→ $(\lambda-0)^2$ → min. 2 free col^ms

$(\lambda-3)$

Q. $(\lambda-2)^3(\lambda-4)(\lambda-5) = 0$

→ $A_{5\times 5}$

$|A| = 2^3 \cdot 4 \cdot 5 = 160$

rank = 5

Q. Given $A_{10\times10}$ with $\overset{\min.}{\wedge}3$ LI eigen vectors, which of the foll^n are possible characteristic eq^{n}s?

a) $(\lambda-1)(\lambda-2)(\lambda-3)=0$

b) $(\lambda-1)(\lambda-2)^9=0$

✓ c) $(\lambda-1)(\lambda-2)^8(\lambda-3)=0$

d) $(\lambda-1)^9(\lambda-2)^3(\lambda-3)=0$

e) $(\lambda-1)(\lambda-2)(\lambda-3)(\lambda-4)^6$

f) $(\lambda-2)^{10}=0$

✓ g) $\lambda^8(\lambda-1)(\lambda-4)=0$

Q. write down properties w.r.t. given char. eq^{n}s.

a) $\lambda^5(\lambda-\lambda_1)^6=0$

$\lambda=0$ is a soln.

$A_{11\times11}$ min 2 LI eig. vecs.

$(\lambda-0)^5$ → max 5 LI eig. vecs.

nullity ≤ 5

$10 \geq$ rank ≥ 6

trace $(A) = 6\lambda_1$ $\Sigma\lambda_i$

det $(A) = 0$ $\Pi\lambda_i$

b) $(\lambda-3)^4(\lambda-4)^5(\lambda-1)=0$

$A_{10\times10}$

rank $= 10$

min LI eigen vectors $= 3$

max LI eig. vecs. $= 10$

Trace $= 4\cdot3+5\cdot4+1$

Det $= 3^4\cdot4^5\cdot1^1$

c) $(\lambda+4)^5(\lambda-4)^3\lambda^2=0$

$A_{10\times10}$

nullity ≤ 2

rank ≥ 8

min 3 LI eig. vectors. max 10

Trace $= (5\cdot(-4))+(3\cdot(4))+(2\cdot0)$

Det $=0$

d) $(\lambda+3)(\lambda+2)(\lambda+1)=0$

$A_{3\times3}$

full rank.

exactly 3 LI eig. vecs.

GM ≤ AM ?

→ $|A-\lambda I| = 0 \Rightarrow \lambda^n + a\lambda^{n-1} + \dots + z = 0$

$A-\lambda I$ has some free variables w.r.t. each value of λ.
↳ we don't know, but max = AM(λ)

(A')

$|A'| = 0$
↳ atleast 1 free var.
will be upper bounded by # of λs.

If $\lambda = 0$:
$A' = A - \lambda I = A$
In echelon form, A will have 1 free var ⇒ 1 zero row.
$\Rightarrow |A| = 0$

If $\lambda = 0$ is repeated k times:
In echelon form, A will have max k free variables.
min 1 free variable.

upper bound (# of free variables) = AM of λ

Rank and nullity when $\lambda = 0$:

even if $\lambda \neq 0$, A' has some free variables (but we do not know about them)

$\lambda = 0$:
multiplicity (AM) of $(\lambda = 0)$ = k
⇒ # of free variables ≤ k
⇒ Nullity ≤ k
⇒ rank ≥ n-k

$n - k \leq \text{rank} \leq n - 1$

If $\lambda = 0$ has multiplicity 1, then A has rank $n-1$.

If an eig. val λ_1 is repeated k times
⇒ atleast 1 LI vec for λ_1
⇒ atmost k LI vecs for λ_1

Real symmetric matrix:

- exactly k LI eig. vecs for λ_i if λ_i is repeated k times.
- eigen vectors are orthogonal to one another.

Cayley-Hamilton theorem

Every square matrix satisfies its own (characteristic) polynomial.

$$\det(A-\lambda I) = (-1)^n \lambda^n + c_{n-1}\lambda^{n-1} + \dots + c_0 = 0$$

$$\Rightarrow (-1)^n A^n + c_{n-1} A^{n-1} + \dots + c_0 = 0$$

E.g. $\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ $\quad \det(A-\lambda I) = \begin{vmatrix} 2-\lambda & 1 \\ 1 & 3-\lambda \end{vmatrix} = \lambda^2 - 5\lambda + 5 = 0$

By CH theorem,

$$A^2 - 5A + 5I = 0$$

$$A^2 = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} 5 & 5 \\ 5 & 4 \end{bmatrix}$$

$$A^2 - 5A + 5I = \begin{bmatrix} 5 & 5 \\ 5 & 4 \end{bmatrix} - \begin{bmatrix} 10 & 5 \\ 5 & 15 \end{bmatrix} + \begin{bmatrix} 5 & 0 \\ 0 & 5 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$$ CH holds.

computing A^{-1} using C-H theorem:

Say characteristic eqn is $\lambda^5 + 3\lambda - 5 = 0$

$$\Rightarrow \lambda^5 + 3\lambda = 5$$
$$\Rightarrow A^5 + 3A = 5I$$
$$\Rightarrow A^4 + 3I = 5A^{-1}$$
$$\Rightarrow A^{-1} = \frac{1}{5}(A^4 + 3I)$$

- characteristic polynomial of AB and BA is the same. $|AB - \lambda_n I_n| = |BA - \lambda_n I_n|$
- AB and BA have the same NON-ZERO eigen values. } same thing

$$(AB)x = \lambda x$$
$$(BA)(Bx) = B\lambda x = (\lambda)(Bx)$$
$$BA(Bx) = \lambda(Bx)$$

(λ: eig. val; Bx: eig. vec)

Say $A_{4\times 3}$.

$B_1 = (AA^T)_{4\times 4}$

$B_2 = (A^TA)_{3\times 3}$

AB and BA share non-zero eigenvalues.

$AA^T \rightarrow$ 4 eig. values
$A^TA \rightarrow$ 3 eigen values
} so extra λ in AA^T is $\lambda = 0$.

$$ABx = 0x$$
$$BAx' = \lambda x'$$

If $\lambda = 0$ is also shared, there's no constraint on x. But we need $x' = Bx$.

- For sq. matrices $A_{n\times n}$ $B_{n\times n}$: same λ's. (zero, non-zero, all.)